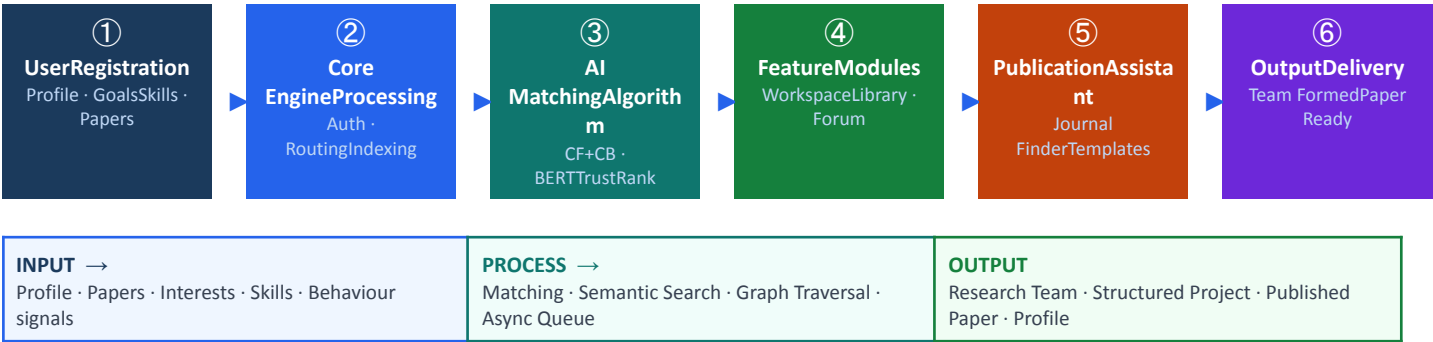


Smart Research Collaboration Platform

Master's Project Project Overview · Mostofa Aminur Rashid · MIT Programme · 2026

1. End-to-End System Flow



2. System Architecture — Layer by Layer

PRESENTATION LAYER	BUSINESS LOGIC LAYER	DATA LAYER
Next.js (React) • Server-Side Rendering (SSR) • Tailwind CSS UI • Socket.IO real-time sync • Cloudflare CDN + DDoS • Responsive dashboard	Node.js + Express (API) • JWT / OAuth 2.0 auth • Rate limiting & routing Python FastAPI (ML) • CF+CB matching engine • Sentence-BERT inference • TrustRank scoring • Kafka async event bus	PostgreSQL Users · Projects · Posts Neo4j (Graph DB) Researcher–Paper–Topic graph Elasticsearch Semantic + full-text search Redis Pre-cached recommendation feed

Step-by-Step Process

#	Action	What the System Does	Key Technical Detail
1	Step 1	User signs up & fills profile	Domain tags, skills, institution, research goals stored in PostgreSQL
2	Step 2	Profile indexed	Sentence-BERT converts profile text → dense vector embedding
3	Step 3	Matching runs	CF+CB hybrid algorithm computes cosine similarity across all profiles
4	Step 4	Feed served	Top matches cached in Redis; user sees collaborators, papers, projects
5	Step 5	Collaboration begins	User joins workspace: task board, doc editor, milestone tracker
6	Step 6	Publication	Publication Assistant queries DOAJ/Scimago for journal recommendations

3. UML Activity Diagram

USER	API / AUTH	ML SERVICE	DATA STORES
Register & fill profile			
↓	↓	↓	↓
◇ ◇ Profile complete?			
↓	↓	↓	↓
Submit profile data	Validate & authenticate JWT		
↓	↓	↓	↓
=== FORK: Parallel Async Processing ===			
	Publish PROFILE_CREATED to Kafka	Consume Kafka event	Receive write request
↓	↓	↓	↓
		Run CF+CB Hybrid Match	Index in Elasticsearch
↓	↓	↓	↓
		Sentence-BERT embedding	Insert node in Neo4j
↓	↓	↓	↓
		◇ Confidence > threshold?	Cache feed in Redis
↓	↓	↓	↓
=== JOIN: Results assembled ===			
↓	↓	↓	↓
View recommendation feed			
↓	↓	↓	↓
◇ ◇ Join project or invite collaborator?			
↓	↓	↓	↓
Select collaborator	Create workspace + notify		Log event to PostgreSQL
↓	↓	↓	↓
Co-author in workspace			
↓	↓	↓	↓
Run Publication Assistant			
↓	↓	↓	↓
	Query DOAJ / Scimago	LLM: generate citation file	
↓	↓	↓	↓
◇ ◇ Checklist complete?			
↓	↓	↓	↓
Submit paper © END		Update researcher profile	Record contribution audit

4. Technology Stack Rationale

Layer	Technology	Why This Choice
Frontend	Next.js (React)	SSR = SEO-indexable academic content; faster page load
UI	Tailwind CSS	Utility-first; zero runtime overhead; rapid iteration
Real-time	Socket.IO	WebSocket + auto-fallback; collaborative editing & live notifications
API Gateway	Node.js + Express	Non-blocking I/O; handles 10k+ concurrent connections efficiently
ML Service	Python + FastAPI	ASGI async; native Python ML ecosystem; auto-generates OpenAPI docs
Matching Algo	CF + CBF Hybrid	CBF solves cold-start; CF improves with user behaviour over time
Semantic Search	Sentence-BERT	Dense embeddings understand meaning, not just keywords
Trust Scoring	TrustRank (PageRank)	Credibility flows from verified professors; resists spam/gaming
Primary DB	PostgreSQL	ACID-compliant; handles complex relational queries; battle-tested
Graph DB	Neo4j	Multi-hop traversal ("2nd-degree collaborators") in milliseconds
Search Index	Elasticsearch	Unified BM25 + kNN vector search across people, papers, posts
Cache	Redis	Pre-computes recommendation feed as background job; <50ms load
Message Queue	Apache Kafka	Decouples API from ML workers; async processing; durable log
Infrastructure	Docker + Kubernetes	Per-service scaling; ML pods scale independently on load spikes

5. Impact Summary — Problem vs. Solution

#	Problem	→ ResearchBridge Solution	Measurable Outcome
1	New researchers don't know who to collaborate with — no structured discovery tool	→ CF+CB Hybrid + Sentence-BERT matching engine surfaces relevant collaborators from profile + behaviour signals	Matched collaborators surfaced within first login; relevance improves with use
2	No shared workspace — researchers use email, Google Drive, WhatsApp ad-hoc	→ Built-in collaboration workspace: task board, milestone tracker, real-time document co-editing	All project activity in one place; version history; no context lost
3	New researchers don't know how to publish — journal selection, formatting, peer review is tacit knowledge	→ Publication Assistant: journal recommender (DOAJ/Scimago) + formatting templates (IEEE/APA/ACM) + pre-submission checklist	Reduces "I don't know where to submit" from weeks of search to minutes
4	Community forums decay into spam — no credibility ranking	→ TrustRank algorithm: PageRank-variant weights credibility by citations + endorsements; professor posts rank higher	Quality-first feed; verified institutional badges (.edu/.ac.bd) auto-assigned
5	Cold-start problem: new users with no history get no useful recommendations	→ Hybrid approach: CBF bootstraps from explicit tags; CF activates as behaviour accumulates	Even 0-paper researchers get relevant results from day one

Performance Targets

Metric	Current (Manual)	With ResearchBridge	How
Find a collaborator	Weeks (conferences, email, luck)	< 5 minutes	Matching engine on first login
Set up shared workspace	1–2 days (tools, permissions, folders)	< 2 minutes	Auto-created on team formation
Find right journal to submit to	Days of research + senior advice	< 10 minutes	Publication Assistant + DOAJ query
Dashboard load time	N/A	< 50 ms	Redis pre-cached feed

Semantic search response	N/A	< 200 ms	Elasticsearch kNN index
Scale to 1M users	Not possible (monolith tools)	Horizontal K8s autoscale	Per-service pod autoscaling

Competitive Gap — What Existing Platforms Are Missing

Feature	ResearchGate	Academia.edu	Overleaf	Semantic Scholar	ResearchBridge
Smart collaborator matching	✗	✗	✗	✗	✓ CF+CB+BERT
Collaboration workspace	✗	✗	✓ writing only	✗	✓ Full (tasks, docs, milestones)
Publication assistant	✗	✗	Partial	✗	✓ Journal finder + templates
Mentorship / community	Partial	✗	✗	✗	✓ TrustRank forum
Built for new researchers	✗	✗	✗	✗	✓ Core design goal
Free to deploy (beta)	—	—	—	—	✓ \$0 on free tiers